

Semi-knockoffs: a model-agnostic Conditional Independence Testing method with finite-sample guarantees

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Summary

Semi-knockoffs is a conditional independence test that works with any pre-trained machine learning model, avoids train–test splitting, and provides p -values and FDR control.

Keywords: Conditional Independence, Scientific Discovery, Knockoffs, Interpretable ML.

Motivation

Which covariates X^j are important with respect to an outcome y ?

$$S = \{j, X^j \not\perp\!\!\!\perp y \mid X^{-j}\}$$

🔍 Study their relationship using a **Machine Learning model**: $\widehat{m} \in \operatorname{argmin}_{f \in \mathcal{F}} \widehat{\mathbb{E}}[\mathcal{L}(f(X), y)]$

Related work: Model-X and Exchangeability under the Null

Model- X assumption: We can sample $\widetilde{X}^{(j)} \sim \mathcal{L}(X^j \mid X^{-j})$.

- Rather than modeling the X - y relationship, we model the input distribution X .
- E.g., when having unlabeled data or measurements from the same device.

Exchangeability under the Null

For any importance statistic $T: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, there is **exchangeability** between

$$T(X^j, X^{-j}, y) \quad \text{and} \quad T(\widetilde{X}^{(j)}, X^{-j}, y)$$

We need **explainable ML** to quantify the contribution of a feature in T

- **Coefficient**-based importance: $T(X^j, X^{-j}, y) = |\widehat{\beta}^j|$
- **Performance**-based importance: $T(X^j, X^{-j}, y) = \ell(\widehat{m}(X^j, X^{-j}), y)$

$$\left\{ \ell(\widehat{m}(X^j, X^{-j}), y) \right\}_{i=1}^{n_{\text{test}}} \quad \text{with} \quad \left\{ \ell(\widehat{m}(\widetilde{X}^{(j)}, X^{-j}), y) \right\}_{i=1}^{n_{\text{test}}}, \quad \widetilde{X}^{(j)} \sim \mathcal{L}(X^j \mid X^{-j}).$$

☹ This requires a train–test split, which significantly reduces the method's performance.

Semi-knockoffs

Semi-knockoffs idea

We propose comparing

$$\left\{ \ell(\widehat{m}(\widetilde{X}_1^{(j)}, X^{-j}), y) \right\}_1^n \quad \text{vs} \quad \left\{ \ell(\widehat{m}(\widetilde{X}_2^{(j)}, X^{-j}), y) \right\}_1^n \quad \text{with} \\ \widetilde{X}_1^{(j)} \sim \mathcal{L}(X^j \mid X^{-j}), \quad \widetilde{X}_2^{(j)} \sim \mathcal{L}(X^j \mid X^{-j}, Y).$$

Key insight: If $X^j \perp\!\!\!\perp y \mid X^{-j}$, $\mathcal{L}(X^j \mid X^{-j}) = \mathcal{L}(X^j \mid X^{-j}, y)$, so they are **exchangeable**.

- We compare **perturbed** vs. **perturbed** instead of **perturbed** vs. **original** data.
- No train–test split is required and works with any model $\widehat{m}$.

No Need to Sample from $\mathcal{L}(X^j \mid X^{-j}, y)$

Problem: Directly sampling $\widetilde{X}_2^{(j)} \sim \mathcal{L}(X^j \mid X^{-j}, y)$ is complex.

Exchangeability under H_0^j :

$$\rho(X^{-j}, y) := \mathbb{E}[X^j \mid X^{-j}, y] = \mathbb{E}[X^j \mid X^{-j}] =: \nu(X^{-j})$$

Idea: Construct new coordinates using *residuals*.

$$\widetilde{X}_1^{(j)} = \nu(X^{-j}) + \underbrace{x_l^j - \nu(x_l^{-j})}_{\epsilon_{j,1}}, \quad \widetilde{X}_2^{(j)} = \rho(X^{-j}, y) + \underbrace{x_k^j - \rho(x_k^{-j}, y_k)}_{\epsilon_{j,2}}$$

- Residuals $\epsilon_{j,1}, \epsilon_{j,2}$ are then exchangeable.
- Avoids conditional sampling.

Type-I error control

Using exchangeability, we obtain type-I error control with any nonparametric paired test.

Algorithm 1 Semi-knockoffs Wilcoxon (SKO-Wcx)

Input: a pretrained model $\widehat{m}$

(Imputers) Fit $\widehat{\nu}_j \approx \mathbb{E}[X^j \mid X^{-j}]$ and $\widehat{\rho}_j \approx \mathbb{E}[X^j \mid X^{-j}, Y]$

(Residuals) Compute $\widehat{\epsilon}_{j,1} = X^j - \widehat{\nu}_j(X^{-j})$ and $\widehat{\epsilon}_{j,2} = X^j - \widehat{\rho}_j(X^{-j}, Y)$

(Permutations) Sample $\pi_{j,1}$ and $\pi_{j,2}$ permutations of $\{1, \dots, n\}$

$$\widetilde{X}_{1,i}^{(j)} = \widehat{\nu}_j(X_i^{-j}) + \widehat{\epsilon}_{j,1,\pi_{j,1}(i)} \quad \text{and} \quad \widetilde{X}_{2,i}^{(j)} = \widehat{\rho}_j(X_i^{-j}, y_i) + \widehat{\epsilon}_{j,2,\pi_{j,2}(i)}$$

Return Wilcoxon test between $\left\{ \ell(\widehat{m}(\widetilde{X}_{1,i}^{(j)}, X_i^{-j}), y_i) \right\}_i$ and $\left\{ \ell(\widehat{m}(\widetilde{X}_{2,i}^{(j)}, X_i^{-j}), y_i) \right\}_i$

False Discovery Rate (FDR) control

For multiple testing, we aim control the

$$\text{FDR}(\widehat{S}) = \mathbb{E} \left[\frac{|\widehat{S} \cap \mathcal{H}_0|}{|\widehat{S} \vee 1|} \right].$$

Knockoffs. If the statistics W_j satisfy the sign-flip property

$$W_j \stackrel{d}{=} \varepsilon_j W_j, \quad \varepsilon_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{\pm 1\}, \quad j \in \mathcal{H}_0,$$

the knockoff threshold provides finite-sample FDR. Our **Semi-knockoff** statistic is

$$W_j^{\text{SKO}} := \frac{1}{n} \sum_{i=1}^n \left[\ell(\widehat{m}(\widetilde{X}_{1,i}^{(j)}, X_i^{-j}), y_i) - \ell(\widehat{m}(\widetilde{X}_{2,i}^{(j)}, X_i^{-j}), y_i) \right],$$

and satisfies the required symmetry, so the standard knockoff threshold can be applied.

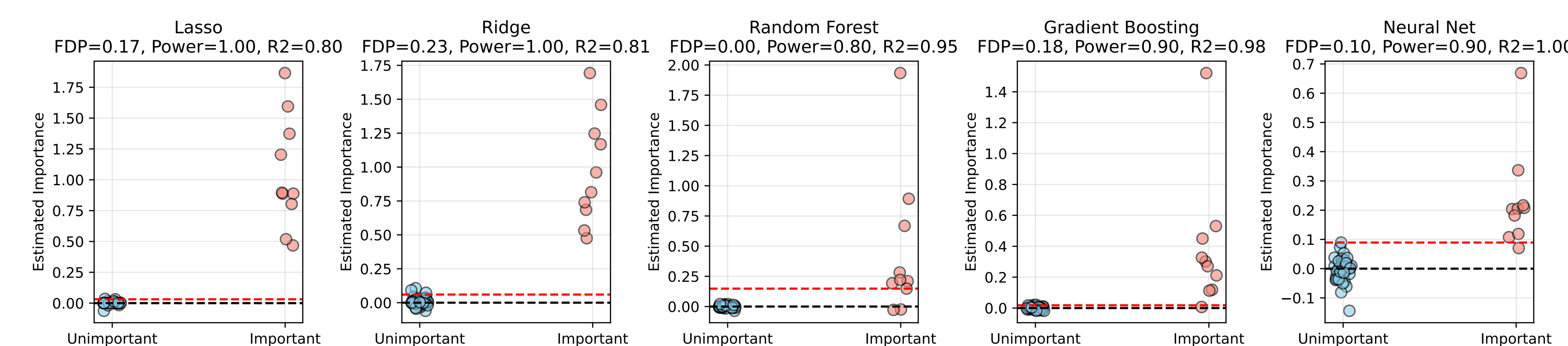


Figure: **Semi-knockoff statistics.** Null features (left) are approximately symmetric around zero.

Experiments

Simulated data: linear setting ($p = 50$, $n = 300$) with a gradient boosting.

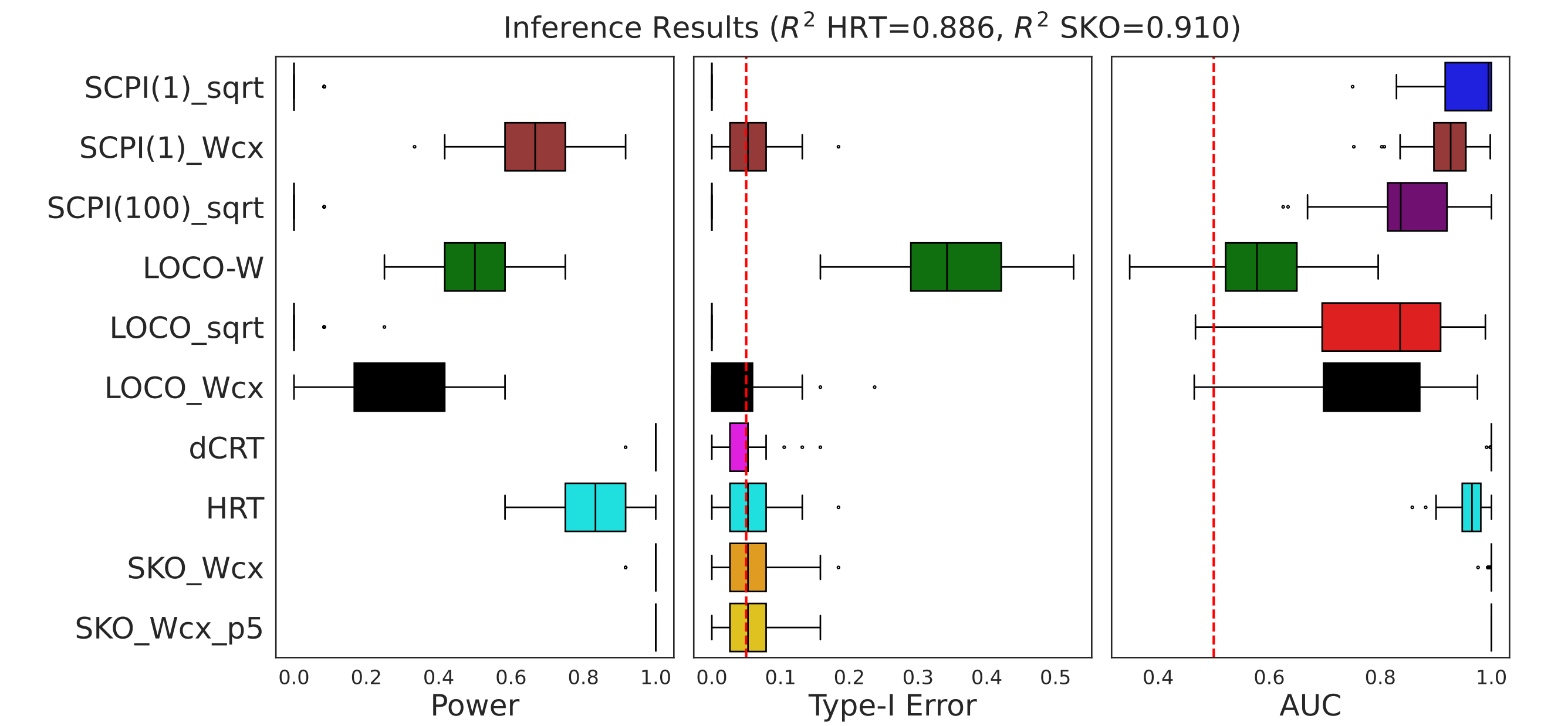


Figure: **SKO controls type-I error with high power by avoiding a train–test split.**

Real data: Type-I error (right of each plot) is estimated under an artificial null.

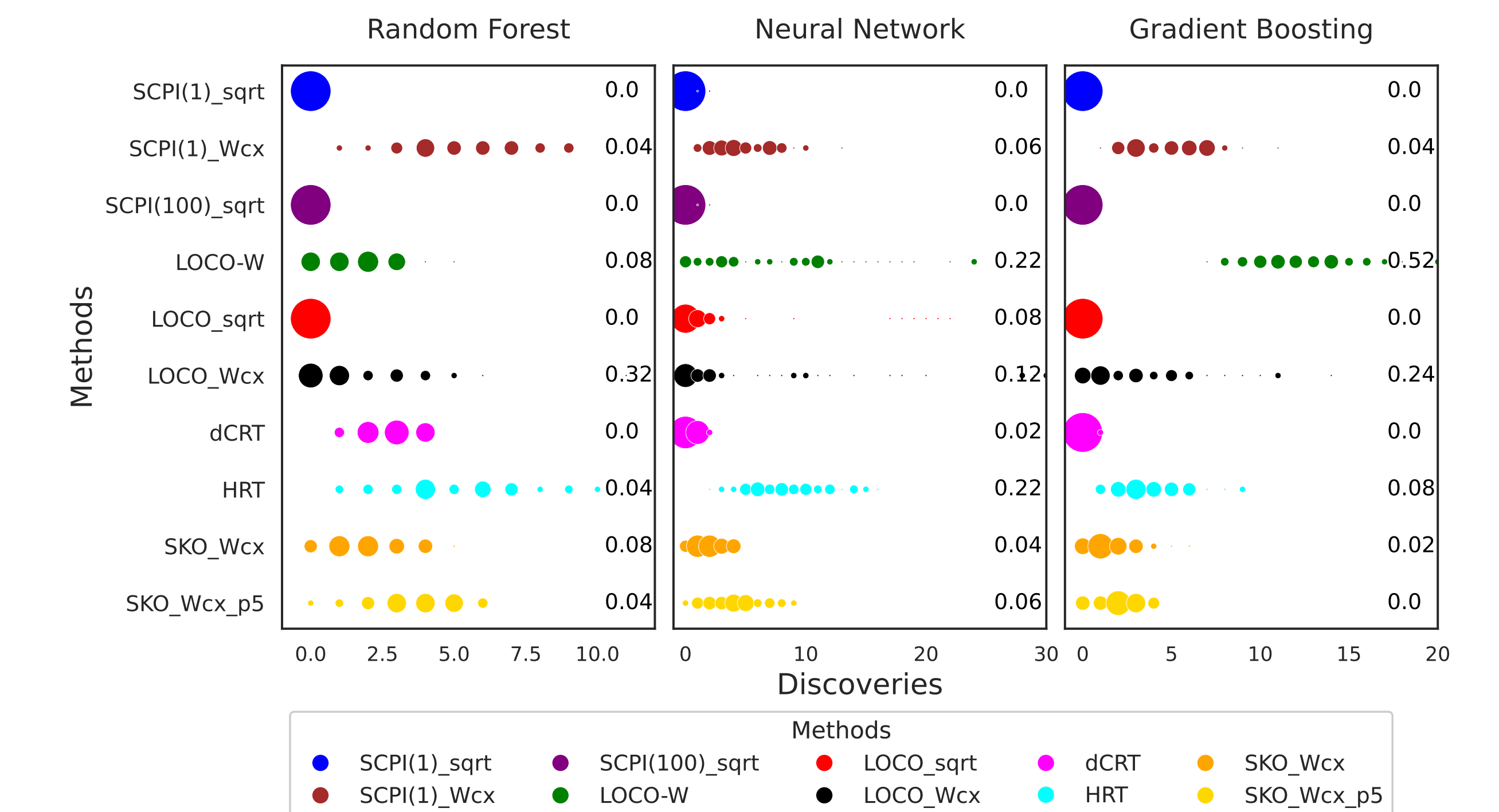


Figure: **Discoveries on real data across ML models.**

Take-home message

- **A flexible conditional independence test** compatible with *any* predictor $\widehat{m}$.
- **Finite-sample guarantees**
 - *Type-I* error control via nonparametric paired tests (e.g., Wilcoxon).
 - *FDR* control via the knockoff threshold.
- **Conditional expectations** provide a practical alternative to densities for modern ML.
- **Robustness** to Model- X misspecification.
- **Theory for estimated imputers** ($\widehat{\nu}_j, \widehat{\rho}_j$)
 - *Stability* of regularized ML models when adding a null feature.
 - *Double robustness* yielding faster convergence rates.



References

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